

EPS 166: Crustal Deformation



FALL 2026

GENERAL INFORMATION

Preamble

Read this syllabus closely, noting the dates of both the activities and when these items are due. If you have questions about our expectations, please contact the professor.

Course

Crustal Deformation

Time

MW 1-2:50

Location

Earth and Physical Sciences 2231

Professor

Amanda M. Thomas

Contact Information

amthom@ucdavis.edu

Office Hours

By appointment

Course Description

This course explores how geoscientists measure and model deformation of Earth's crust to understand earthquakes, slow slip, volcanoes, and other geologic processes. We use GNSS, InSAR, seismic, and other observations to connect surface deformation to processes occurring at depth, with an emphasis on quantitative reasoning, physical modeling, inverse problems, and interpretation of real geophysical data.

Textbook

This course has no assigned textbook. Relevant readings will be provided by the instructor.

Learning Outcomes

By the end of the quarter, you will be able to:

- Explain how crustal deformation reflects physical processes including fault slip, earthquakes, magma movement, and fluid withdrawal.
- Interpret GNSS, InSAR, seismic, and other geophysical observations to characterize the spatial and temporal patterns of crustal deformation.
- Apply elastic and kinematic models to predict surface deformation produced by subsurface sources.
- Use inverse methods to estimate fault slip and other source properties from geodetic observations, and evaluate nonuniqueness, uncertainty, and model assumptions.
- Integrate multiple geophysical datasets to make quantitative inferences about earthquakes, the earthquake cycle, slow slip, volcanic unrest, and other crustal processes.

COURSE SCHEDULE

Week	Topic	Dates
Week 0	A Deforming Earth	9/23
Week 1	GNSS Positioning and Timeseries	9/28 and 9/30
Week 2	Strain and the Seismic Cycle	10/5 and 10/7
Week 3	Short and long timescale Seismology	10/12 and 10/14
Week 4	InSAR	10/19 and 10/21
Week 5	Forward and Inverse Modeling	10/26 and 10/28

Week 6	Midterm and Slip Inversion	11/2 and 11/4
Week 7	Mass Movements	11/9
Week 8	Volcano Deformation and Magma Migration	11/16 and 11/18
Week 9	Groundwater, Fluids, and Subsidence	11/23; no class 11/25
Week 10	Vertical Land Motion and Synthesis	11/30 and 12/2

ASSESSMENT AND GRADING

Grading

Grades in this course are assigned with the following weights:

1. Lab Notebooks - 40%
2. Midterm - 15%
3. Final - 15%
4. Attendance - 15%
5. Participation - 15%

We maintain a record of your grades in this course using Canvas. Please check Canvas often to ensure that the grade recorded is the one you received (mistakes happen!).

In-class Lab Notebooks

In-class Jupyter notebook laboratories provide guided, hands-on exercises that reinforce lecture concepts through exploration and visualization. These activities emphasize building physical intuition, connecting theory to observations, and working with simple analysis tools.

Each question within a lab will receive one of the following marks:

- **Check-plus:** The response is complete, correct, and demonstrates thoughtful interpretation or reasoning.
- **Check:** The response demonstrates a reasonable attempt and general understanding but contains minor errors or is not fully developed.
- **No check:** The response is missing, substantially incomplete, or does not meaningfully address the question.

For calculating the lab grade, a check-plus is worth 100%, a check is worth 85%, and no check is worth 0%. The overall lab grade will be the average of the scores received on its individual questions.

Notebooks should be uploaded to Canvas no later than **one week** after the material has been presented in class. For example, if we cover material in a notebook on Wednesday, the completed notebook should be uploaded to Canvas no later than the start of class on Wednesday of the following week.

Midterm and Final Exams

The midterm and final will be individual oral examinations designed to evaluate your understanding of key concepts and your ability to apply them in quantitative and interpretive contexts. Each exam will last approximately 15 minutes and will generally include three questions: a conceptual explanation, an interpretation of data or a figure, and an application or modeling question. Follow-up questions may be used to clarify your reasoning.

Questions will vary among students but will be drawn from the same topic areas and assessed using the same criteria. Sample questions and topic categories will be provided in advance. You may use scratch paper and draw diagrams, and you may ask for a question to be repeated or clarified. Grading will be based on scientific understanding and reasoning rather than speaking style.

Each question will be scored as follows:

- **3 – Strong:** Accurate, clear, and supported by appropriate reasoning.
- **2 – Satisfactory:** Mostly accurate, with minor errors or omissions.

- **1 – Developing:** Demonstrates partial understanding but includes important errors or requires substantial prompting.
- **0 – Insufficient:** Does not demonstrate meaningful understanding of the concept.

The midterm is scheduled for Monday of Week 6. The final will take place during finals week. Students will sign up for individual examination times.

Attendance

Regular attendance is important because the course covers a substantial amount of material through in-class instruction and activities. Attendance will be taken at the beginning of each class, and you must be present at that time to receive attendance credit. At the end of the quarter, your four lowest attendance scores will be dropped. Therefore, you do not need to notify me about occasional absences; additional absences beyond the four dropped scores will affect your grade. Regardless of attendance, you remain responsible for completing the associated laboratory exercises.

Participation

Participation means actively engaging with the course during class. This may include contributing to discussions, asking questions, taking notes, working through the Jupyter notebooks, participating in group activities, or listening attentively. You do not need to speak frequently to earn full participation credit, but physical presence alone is not sufficient. Participation will be assessed through consistent, observable engagement. Activities unrelated to the course may result in reduced participation credit for that class period. If a laptop or tablet becomes a persistent distraction, I may ask you to put it away.

COURSE POLICIES AND STUDENT SUPPORT

Academic Integrity

You are expected to uphold the [UC Davis Code of Academic Conduct](#). Suspected violations are reported through the official process. Use of generative AI is not permitted on any assignments in this course. Suspected misconduct in this realm will be reported.

Electronic Etiquette

I expect email messages to conform to standards of professional correspondence. Emails should be courteous and respectful in tone; do not send emails that are curt or demanding. Please do not expect an immediate response via email (normally, a response will be sent within one business day). If your question is sent at the last minute, it may not be possible to send you a response before a deadline. Please do not send more than one email per week. If you find that you have more questions than that, make an appointment to speak with me or come talk with me after class.

Laptops/tablets are welcome for learning tasks. Please avoid unrelated browsing and cell phone use during class.

Classroom Climate

Our class follows the UC Davis [Principles of Community](#). Expect a respectful, inclusive environment—questions and engagement are encouraged!

Accessibility and Accommodation

UC Davis is committed to equal access. If you have, or think you may have, a disability that affects learning or participation, please contact the Student Disability Center to obtain a Letter of Accommodation. This letter will be shared with me and allows me to implement accommodations.

Title IX / Responsible Employee Disclosure

I am a Responsible Employee. This means if you share information about sexual violence, sexual harassment, or related misconduct, I must report it to the campus HDAPP/Title IX office, which will

reach out to offer resources and options. If you want confidential support, please consider [CARE](#) or [Student Health and Counseling Services](#).

Wellness and Basic Needs

College is demanding. If you are experiencing food, housing, or financial insecurity, or other basic-needs barriers, please connect with the [Aggie Compass Basic Needs Center](#). For mental health support, [Student Health and Counseling Services](#) offers crisis support and short-term counseling.

Syllabus Changes

The syllabus is a live document; I may make updates as the quarter progresses. I will announce any changes on Canvas.